

TD 7

Exercice 1 – RKHS

Recalls : A vector space E over $\mathbb{R}$ is a set of elements such that it is possible to perform linear combinations of its elements (E is equipped with an addition operation and a scalar multiplication operation);

A function $Q : E \times E \rightarrow \mathbb{R}$ is an inner product if and only if :

1. it is symmetric : $Q(x, y) = Q(y, x)$;
2. it is bilinear : $Q(\lambda_1 x_1 + \lambda_2 x_2, y) = \lambda_1 Q(x_1, y) + \lambda_2 Q(x_2, y)$;
3. it is positive : $Q(x, x) \geq 0$ and $Q(x, x) = 0 \iff x = 0$.

We will often denote $Q(x, y) = \langle x, y \rangle_E$, and the norm associated with an inner product as $\|x\|_Q = \sqrt{Q(x, x)}$; A Hilbert space is a complete vector space equipped with an inner product; A kernel is a function $k : X \times X \rightarrow \mathbb{R}$ such that there exists a Hilbert space $\mathcal{H}$ and a function (projection or *feature map*) $\phi : X \rightarrow \mathcal{H}$ such that $\forall x, x' \in X, k(x, x') = \langle \phi(x), \phi(x') \rangle_{\mathcal{H}}$.

Let $x_1, \dots, x_n \in X$ and a function $k : X \times X \rightarrow \mathbb{R}$, the Gram matrix of k is the matrix $\mathcal{K} := k_{i,j} = k(x_i, x_j)$. A matrix is said to be positive semidefinite if $\forall c_i \in \mathbb{R}, \sum_{i,j} c_i c_j k_{i,j} \geq 0$. In this case, the function is also said to be positive definite.

Q 1.1 Express $\sum_{i,j} c_i c_j k_{i,j}$ as an inner product. Show that a kernel is positive definite.

$$\begin{aligned} \sum_{i,j} c_i c_j k_{i,j} &= \sum_{i,j} c_i c_j \langle \phi(x_i), \phi(x_j) \rangle \text{ since } k \text{ is an admissible kernel.} \\ \sum_{i,j} c_i c_j \langle \phi(x_i), \phi(x_j) \rangle &= \sum_{i,j} \langle c_i \phi(x_i), c_j \phi(x_j) \rangle = \langle \sum_i c_i \phi(x_i), \sum_j c_j \phi(x_j) \rangle = \left\| \sum_i c_i \phi(x_i) \right\|^2 \geq 0 \end{aligned}$$

Q 1.2 The goal of this question is to prove the contrapositive, that a symmetric positive semidefinite function $k : X \times X \rightarrow \mathbb{R}$ is a kernel. To do so, we need to find a Hilbert space $\mathcal{H}$, an inner product $Q : \mathcal{H} \times \mathcal{H} \rightarrow \mathbb{R}$ and a projection $\phi : X \rightarrow \mathcal{H}$ such that $k(x, y) = Q(\phi(x), \phi(y)) \quad \forall x, y \in X$. We consider $\mathcal{H}$ as the vector space spanned by functions of the form $y \mapsto k(y, x)$ for all $x \in X$. An element of $\mathcal{H}$ is therefore a function from $X \rightarrow \mathbb{R}$.

Let $\Phi : X \rightarrow \mathcal{H} := k(\cdot, x)$ be a mapping from X to the functions of $\mathcal{H}$, where $\Phi(x)(x') = k(x', x)$. Let $\alpha_i \in \mathbb{R}, \beta_i \in \mathbb{R}, x_i \in X, x'_i \in X$ for $i \in \{1..n\}$. We define :

$$f(\cdot) = \sum_{i=1}^n \alpha_i \Phi(x_i)(\cdot), \quad g(\cdot) = \sum_{i=1}^n \beta_i \Phi(x'_i)(\cdot), \quad Q(f, g) = \sum_{i=1}^n \sum_{j=1}^n \alpha_i \beta_j k(x_i, x'_j)$$

f and g belong to $\mathcal{H}$ since they are linear combinations of elements in $\mathcal{H}$.

Q 1.2.1 Show that Q can be expressed using only β_j and $f(x'_j)$, or α_i and $g(x_i)$.

We define f and g in this way because it allows for generality and we can use them to demonstrate that Q , which is assumed to be positive semidefinite, is an inner product of f and g . This will allow us to decompose k as an inner product of a projection in $\mathcal{H}$, thus proving that k is a kernel.

$$\begin{aligned} Q(f, g) &= \sum_{i=1}^n \sum_{j=1}^n \alpha_i \beta_j k(x_i, x'_j) = \sum_{i=1}^n \sum_{j=1}^n \alpha_i \beta_j \Phi(x'_j)(x_i) = \sum_{i=1}^n \alpha_i g(x_i) \\ Q(f, g) &= \sum_{i=1}^n \sum_{j=1}^n \alpha_i \beta_j k(x'_j, x_i) \text{ since } k \text{ is symmetric.} \end{aligned}$$

$$\text{Thus, } Q(f, g) = \sum_{i=1}^n \sum_{j=1}^n \alpha_i \beta_j \Phi(x_i)(x'_j) = \sum_{i=1}^n \beta_i f(x'_i)$$

Q 1.2.2 Show that $Q(f, g)$ is an inner product of f and g (we can then replace $Q(f, g)$ with $\langle f, g \rangle$). To do this, prove that :

- Q is symmetric.

$$\text{Since } k \text{ is symmetric, then } Q(g, f) = \sum_{i=1}^n \sum_{j=1}^n \beta_i \alpha_j k(x'_i, x_j) = \sum_{i=1}^n \sum_{j=1}^n \alpha_i \beta_j k(x_i, x'_j) = Q(f, g)$$

- Q is bilinear.

$$\begin{aligned} Q(\lambda f, g) &= \sum_{i=1}^n \beta_i \lambda f(x'_i) = \lambda \sum_{i=1}^n \beta_i f(x'_i) = \lambda Q(f, g) \\ Q(f + f', g) &= \sum_{i=1}^n \beta_i (f(x'_i) + f'(x'_i)) = \sum_{i=1}^n \beta_i f(x'_i) + \sum_{i=1}^n \beta_i f'(x'_i) = Q(f, g) + Q(f', g) \end{aligned}$$

- $Q(f, f) \geq 0$ (we will show in the last question that $Q(f, f) = 0 \iff f = 0$).

$$Q(f, f) = \sum_{i=1}^n \alpha_i f(x_i) = \sum_{i=1}^n \sum_{j=1}^n \alpha_i \alpha_j \Phi(x_j)(x_i) = \sum_{i=1}^n \sum_{j=1}^n \alpha_i \alpha_j k(x_i, x_j) \geq 0 \text{ (since } k \text{ is positive semidefinite).}$$

Q 1.2.3 What are $Q(k(\cdot, x), f)$ and $Q(k(\cdot, x), k(\cdot, x'))$? Justify the name of k : *reproducing kernel*.

$$\text{Let } g(\cdot) = \sum_{i=1}^n \frac{1}{n} k(\cdot, x) = k(\cdot, x), \text{ then } Q(k(\cdot, x), f) = \frac{1}{n} \sum_{i,j} \alpha_i k(x_i, x) = \sum_i \alpha_i k(x, x_i) = f(x) \\ Q(k(\cdot, x), k(\cdot, x')) = k(x, x')$$

Q 1.2.4 Assuming $Q(f, g)^2 \leq Q(f, f)Q(g, g)$, show that $|f(x)|^2 \leq k(x, x)Q(f, f)$. Conclude.

$$|f(x)|^2 = Q(k(\cdot, x), f)^2 \leq Q(k(\cdot, x), k(\cdot, x))Q(f, f) \leq k(x, x)Q(f, f) \\ \text{Thus, if } Q(f, f) = 0, \text{ then } f(x) = 0 \forall x.$$

Exercise 2 (6 points) – Collaborative Filtering

The goal of this exercise is to propose a model for recommendation (or collaborative filtering) : the aim is to predict a user's opinion on an item (which could be a movie, a product, a restaurant, etc.) based on known ratings between n users (set $\mathcal{U}$) and m items (set $\mathcal{V}$). The ratings (or scores) that users have given to items are represented by a rating matrix R of size $n \times m$, where r_{ij} is the rating of user i for item j . All ratings are assumed to be strictly positive (e.g., between 1 and 5) ; a value of zero for r_{ij} means that the rating of user i for item j is unknown.

A common modeling approach in this context is to consider a *latent space*—not explicit but inferred from the data—of explanatory factors for users' opinions on items. For instance, in the case of movie recommendations, one can imagine that this latent space consists of different movie genres (comedy, thriller, romance, etc.). A movie can then be represented by a set of weights across the different genres—its weight vector—indicating its affinity for each genre. Similarly, a user can be represented by their weight vector across the genres. The predicted score for a user on a movie is then the dot product between the two weight vectors. The least squares error between the predicted score and the observed score is generally used as the cost function.

Q 2.1 Assume a latent space of dimension 4 : (comedy, romance, thriller, crime) and the following data :

	Comedy	Romance	Thriller	Crime
Movies				
v_1 : Angry Bird	0.4	0.2	0.	0.
v_2 : The Big Lebowski	0.8	0.2	0.6	0.4
v_3 : Fargo	0.4	0.1	0.3	0.2
v_4 : Saw	0	0	0.6	0.1
Users				
u_1	0.5	0.5	0.5	0.5
u_2	0.2	0.2	0.6	0.8
u_3	0.1	0.1	0.3	0.8

User	Movie	Score
u_1	v_1	3
u_1	v_3	10
u_2	v_2	9
u_2	v_3	4
u_3	v_3	3
u_3	v_4	3

Q 2.1.1 Compute the score matrix R for these data.

Q 2.1.2 Compare the obtained scores with the given supervision scores, and compute the cost of this approximation. What elementary operation can greatly improve the cost ? Is it important ?

Q 2.1.3 What do you notice about the scores and the representation of movies v_2 and v_3 ? Are they close or far apart ? Explain the observed phenomenon, considering the average ratings for each movie. Does the Euclidean distance seem suitable for calculating proximity between movies (or between users) ? Which other distance could be used ?

Q 2.2 Let R be the score matrix of size $n \times m$, with each row r_i corresponding to the scores given by a user and each column r_j to the scores received by an item. Assume the entire matrix is known. Formalize the recommendation problem described in the statement as the search for two matrices U and V of sizes $n \times d$ and $d \times m$ that minimize an expression involving R . What do U , V , and d represent ?

Q 2.3 To the term found in the previous question, we add the term $\lambda(\sum_i \|u_{i,\cdot}\|^2 + \sum_j \|v_{\cdot,j}\|^2)$ to establish the final quantity to optimize. What do the sums in this second term represent ? What is their purpose ?

Q 2.4 In practice, we rarely have access to all the scores : the matrix R is sparse, with very few known values. How can the optimization problem be adapted ?

Q 2.5 The problem is not directly optimizable : why ? One solution is to alternate the optimization of the two matrices U and V : at one step, U is optimized with V fixed, and at the next step, V is optimized with U fixed.

Q 2.5.1 To which classical family of statistical learning problems does each of the two optimization phases correspond ?

Q 2.5.2 For a rating $r_{i,j}$, provide the associated update formulas. Give the complete optimization algorithm (considering stochastic optimization).

Q 2.5.3 Provide the Python implementation.

Q 2.6 Not all users have the same average opinion (a user may be more critical than another but have similar tastes), and the same applies to items. What is missing from the model to account for these biases ? Propose modifications to the algorithm.

1. $[0.3, 1, 0.5, 0.35]$, $[0.12, 0.88, 0.44, 0.44]$, $[0.06, 0.6, 0.3, 0.26]$, scores are close to within a factor of 10 ; simply multiply one matrix by 10 ; double the score for one, movies are similar, cosine distance instead of Euclidean.
2. $\min_{U,V} \|R - UV\|$, d is the dimension of the latent space, U is the user representation matrix, V is the movie representation matrix.

3. Regularization on users and movies.
4. Ignore unknown values ; only minimize the cost over known values.
5. Linear regression.
6. Gradient descent... to be calculated :)
7. Python.
8. Introduce a bias for each user and a bias for each movie.